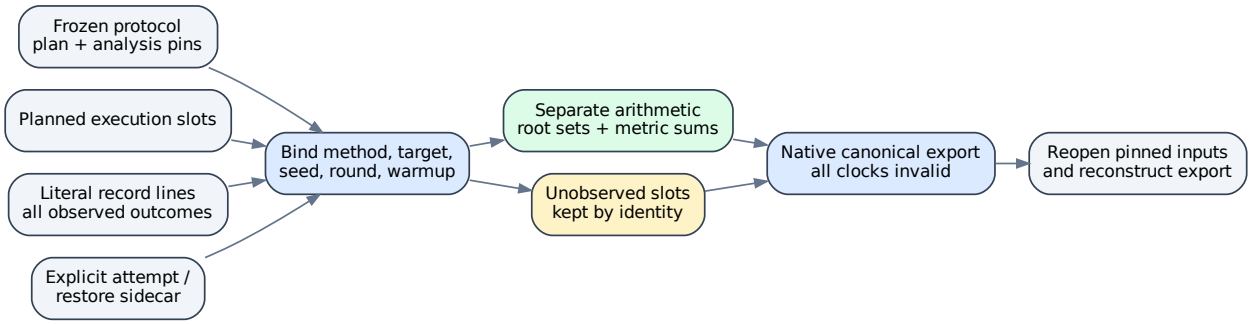
APS native export and bounded audit reader

2026-11-04 — component development report

The Rust adapter constructs the canonical experiment export from a pinned protocol, saved plan, literal record lines and explicit execution-history sidecar. The reader uses separate ordinary-residue affine arithmetic and source reconstruction. All development assertions here have producer standing `[pcc:unverified]`; no signed execution result is supplied.



Collection and consistency. Rows retain failures, late witnesses and recordless infrastructure outcomes. Every unobserved planned slot remains visible by identity, without an invented attempt or exclusion. Checking reconstructs the complete export from pinned input bytes; an incomplete observed population cannot pass the complete check.

Bounded mathematical and metric reader. The explicit-curve guard covers prime fields up to 65,521. Checks include subgroup membership, labeled points, signed coefficients, nonzero denominator and recovered scalar; cubic pruning additionally requires the map/eigenvalue and propagated labels. A direct point oracle compares attempts claiming complete root sets, capped at one million sign candidates per record. Partial root sets remain partial. Six cold phases and five online phases are recomputed, along with the producer's retained partial diagnostics.

Producer-reported development checks	Outcome
Standalone release library	42 passed, 0 failed
Native command-line pipeline	5 passed, 0 failed
Strict Clippy: Rust 1.98 and stable	exit 0 each
Workflow syntax	exit 0
Actual producer bridge	3 passed, 0 failed
Broad repository library	running; failures observed

Claim `PCC-20261104-56667f`: unverified. Normalized commands, source pins and original-log digests are in `validation/source-freeze-summary.json`. The first draft's invalid one-arm fixture and the bridge's corrected crate-name import are retained as development failures. These counts are not timings or experimental samples. Final protocol fixtures are wholly synthetic; copied historical runtime metadata was rejected for publication and replaced. Source commit `a6630cd83e25` pins the component.

Explicit boundaries. Packaging does not qualify a clock or attest execution, sampler history, complete native work, polynomial-construction fidelity or full curve order. All exported clocks remain invalid. Next steps are complete operation pricing, the held-out four-size protocol/analysis and external comparison, followed by maintained run admission. The approved source checkpoint awaits configured authentication. No performance session or benchmark row is reported. Publication of the companion native ledger is blocked pending explicit approval; merging remains disabled.

Sources. APS revised manuscript §§3.1–3.4; prior APS four-lead and replay contracts; canonical `experiment.protocol/v1` and `experiment.export/v1` at crypto-autoresearcher `774611edb0a5`; the native pencil exporter's study-specific reconstruction contract; repository PCC and measurement rules.